15. Decision Tree Classifier

```
from sklearn.tree import DecisionTreeClassifier

# Dataset: [[Weather (1: Sunny, 0: Rainy), Temp (Celsius)]]
X = [[1, 30], [1, 24], [0, 12], [0, 18], [1, 15], [0, 28]]
# Targets: [1: Play Outside, 0: Stay Indoors]
y = [1, 1, 0, 0, 1, 0]

if __name__ == "__main__":
    # Create the model using Entropy (Information Gain)
    classifier = DecisionTreeClassifier(criterion='entropy', random_state=42)
    classifier.fit(X, y)

    print("Decision Tree Model successfully trained.\n")

    # Run categorical sample predictions
    test_cases = [[1, 26], [0, 14]]
    for case in test_cases:
        prediction = classifier.predict([case])[0]
        outcome = "Play Outside" if prediction == 1 else "Stay Indoors"
        print(f"Features: Weather={case[0]}, Temp={case[1]}C -> Decision: {outcome}")
```

Sample Output

text

```
Decision Tree Model successfully trained.

Features: Weather=Sunny, Temp=26°C -> Decision: Play Outside
Features: Weather=Rainy, Temp=14°C -> Decision: Stay Indoors
```